

RASEED - PRODUCT REQUIREMENTS DOCUMENT (PRD)

Version

1.0

Product Name

Raseed

Tagline

Know Where Your Money Goes. Shape Where It Goes Next.

1. Executive Summary

Raseed is an AI-powered Financial Copilot designed for young adults aged 18-30 who struggle with budgeting, saving, and achieving financial goals.

Unlike traditional expense trackers that merely display transaction history and charts, Raseed aims to transform financial data into actionable decisions.

The platform will help users:

- Understand spending behavior
- Track savings goals
- Detect unnecessary expenses
- Forecast future finances
- Receive personalized financial guidance
- Learn about relevant government-backed financial schemes

The project will be developed incrementally through multiple versions, each adding measurable functionality while maintaining a deployable and production-ready system.

2. Problem Statement

Young adults today:

- Frequently use UPI and digital payments
- Have limited awareness of spending patterns
- Lack financial planning discipline
- Do not know how much they should save monthly

- Often fail to connect daily spending habits with long-term goals

Existing finance applications focus on transaction tracking but rarely answer questions such as:

- Can I afford this purchase?
- Will I reach my savings goal?
- Why am I unable to save money?
- Which spending habits are hurting my finances?

Raseed aims to bridge this gap.

3. Target Audience

Primary Users:

- College Students
- Young Professionals
- Freelancers
- Early Career Employees

Age Range:

18-30 Years

Region:

Initially India

Financial Profile:

- Beginner to intermediate financial literacy
 - Monthly income ranging from ₹5,000 to ₹100,000+
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4. Product Vision

To become a personal financial copilot that continuously analyzes user spending patterns and provides intelligent recommendations that improve financial outcomes.

5. Core Product Principles

1. Simplicity First
2. Explainable Insights
3. Trust and Transparency
4. Action-Oriented Recommendations
5. Gradual Automation

6. High-Level Architecture

React Frontend ↓ FastAPI Backend ↓ PostgreSQL Database

Future Layers:

Email Integration Forecasting Engine Vector Database RAG Layer AI Assistant

VERSION 1

Financial Snapshot MVP

Timeline: 2 Months

Objective: Validate that users find value in spending insights and goal tracking.

V1 Features

Authentication

- Register
- Login
- Logout

Authentication Method:

- JWT Tokens
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Statement Upload

Supported Formats:

- PDF

Scope:

- Initially support a single bank format

Examples:

- SBI OR

- HDFC

Only one bank must be supported initially.

PDF Parsing

Pipeline:

PDF ↓ Text Extraction ↓ Transaction Extraction ↓ JSON Conversion

Output Fields:

- Date
 - Merchant
 - Amount
-

Rule-Based Categorization

Categories:

- Food
- Shopping
- Travel
- Entertainment
- Education
- Bills
- Miscellaneous

Example Rules:

SWIGGY → Food

ZOMATO → Food

UBER → Travel

NETFLIX → Entertainment

Implementation:

Keyword-Based Classification

Dashboard

Metrics:

- Total Spending
- Category Breakdown
- Monthly Spending

Visualizations:

- Pie Chart
 - Summary Cards
 - Recent Transactions Table
-

Goal Tracking

User Inputs:

- Goal Name
- Target Amount
- Target Date

Output:

Required Monthly Savings

V1 Database Schema

Users

- id
- name
- email
- password_hash

Transactions

- id
- user_id
- date
- merchant
- amount
- category

Goals

- id
- user_id
- goal_name

- target_amount
 - target_date
-

V1 Tech Stack

Frontend:

- React
- TailwindCSS
- React Router
- Axios

Backend:

- FastAPI
- SQLAlchemy

Database:

- PostgreSQL

Deployment:

- Vercel
 - Render
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VERSION 2

Smart Expense Intelligence

Timeline: Months 3-4

Objective: Reduce manual effort and improve transaction understanding.

V2 Features

Gmail Integration

OAuth Authentication

Read:

- Transaction Alerts
- Bank Notifications
- Purchase Receipts

Supported Providers:

- Gmail
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Merchant Recognition Engine

Examples:

SWIGGY → Food

AMAZON → Shopping

UBER → Travel

Purpose:

Reduce Unknown Transactions

Transaction Type Classification

Transaction Types:

- Income
 - Expense
 - Transfer
 - Investment
 - Loan Payment
-

Spending Trend Detection

Examples:

Food spending increased 20%.

Entertainment spending exceeds average.

Personalized Budget Suggestions

Generate suggested monthly category budgets.

V2 New Tables

EmailAccounts

- id
- user_id
- provider

MerchantMappings

- merchant_name
 - category
-

V2 Technical Additions

OAuth2

Background Tasks

Email Processing Service

Merchant Classification Service

Dockerization

VERSION 3

Financial Planning Engine

Timeline: Months 5-6

Objective: Shift from analytics to financial decision-making.

V3 Features

Cash Flow Forecasting

Predict:

- Month-End Balance
 - Expected Savings
-

Goal Achievement Prediction

Example:

At current spending patterns, goal completion will be delayed by 2 months.

Savings Leak Detection

Examples:

Excessive Food Spending

Subscription Waste

Recurring Purchases

What-If Simulator

Examples:

What if I save ₹2,000 more?

What if I reduce food spending by 15%?

V3 Machine Learning

Forecasting Models:

- Linear Regression
- Prophet
- XGBoost

Initial implementation should prioritize simplicity.

DevOps Additions

Docker Compose

GitHub Actions

Automated Testing

CI/CD Pipeline

Health Monitoring

VERSION 4

AI Financial Copilot

Timeline: Months 7-8

Objective: Enable conversational financial assistance.

V4 Features

Natural Language Queries

Examples:

Where did my money go last month?

Can I afford a bike?

Why am I overspending?

AI Insight Generation

Instead of static charts:

Generate personalized summaries.

Financial Health Score

Factors:

- Savings Rate
 - Spending Discipline
 - Goal Progress
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AI Layer

LLM Integration

Prompt Engineering

Context Generation

VERSION 5

RAG & Knowledge Engine

Timeline: Months 9-10

Objective: Provide grounded financial education and recommendations.

V5 Features

Government Scheme Recommendations

Examples:

- PPF
- NPS
- Sukanya Samriddhi

Educational Financial Guidance

Tax Saving Information

Emergency Fund Guidance

RAG Architecture

User Query ↓ Embedding Generation ↓ Vector Search ↓ Context Retrieval ↓ LLM Response

Knowledge Sources

Government Scheme Documents

Financial Literacy Content

Investment Education Material

Vector Database

Preferred:

PostgreSQL + pgvector

Alternative:

Qdrant

Embedding Models

OpenAI Embeddings

Sentence Transformers

VERSION 6

Industry-Grade Platform

Timeline: Months 11-12

Objective: Improve scale, reliability, and intelligence.

Features

Multi-Email Support

Advanced User Profiling

Reimbursement Detection

Net Worth Tracking

Advanced Analytics

Admin Dashboard

DevOps Expansion

Monitoring

Logging

Observability

Container Orchestration

Security Requirements

Password Hashing:

bcrypt

JWT Authentication

HTTPS Everywhere

Encrypted Environment Variables

No Storage of Banking Credentials

Principle of Least Privilege

Success Metrics

V1

- Upload Success Rate > 90%
- Categorization Accuracy > 80%

V2

- Email Parsing Accuracy > 85%

V3

- Forecast Accuracy Improvement

V4

- Meaningful User Queries Answered

V5

- Accurate Retrieval-Augmented Responses
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Resume Project Description

Built Raseed, an AI-powered Financial Copilot using React, FastAPI, PostgreSQL, and Machine Learning. Developed automated transaction extraction, rule-based categorization, goal-based financial planning,

forecasting models, Retrieval-Augmented Generation (RAG), vector database integration, and CI/CD-enabled cloud deployment.